Dongting Zhai Email : dongting.24@intl.zju.edu.cn Mobile : +86 158 6716 1227

Website: https://iris1618.github.io

EDUCATION

Zhejiang University-University of Edinburgh Institute (ZJE) **Sept. 2024 - Present**

- Undergraduate in Biomedical Informatics, Year 3
- cGPA: 3.95/4.30
- Expected Graduation: Jun 2028

RESEARCH EXPERIENCES

Rui Lin Lab, National Institute of Biological Sciences (NIBS), Beijing
Summer Research Intern, Neuroscience | Jun 2026– Aug 2026

Project 1: Whole-Brain Microglial Gene Editing

- Explored whole-brain microglial gene editing through primary microglia transplantation into adult mice.
- Performed primary microglia isolation and culture, AAV-mediated gene editing, oral gavage, and stereotaxic injections.
- Conducted mouse perfusion, brain sectioning, immunofluorescence staining, and flow cytometry to evaluate microglial engraftment.

Project 2: CD39 Regulation of Injury-Induced ATP Signaling

- Investigated the role of CD39 in injury-induced ATP signaling using CD39 overexpression and in vivo two-photon imaging.
- Performed stereotaxic AAV injections and cranial window surgery, and gained hands-on experience in two-photon microscopy.

Wanlu Liu Lab, Zhejiang University–University of Edinburgh Institute (ZJE)
University-level Student Research Training Program (SRTP) — Spatial Transcriptomics
Apr 2026 – Present

- Analyzed Visium HD spatial transcriptomics data from liver samples with different disease backgrounds.
- Performed data preprocessing, cell type annotation, differential expression analysis, and pseudo-spatial analysis.
- Aim to investigate hepatocyte zonation along the portal vein–central vein (PV–CV) axis and spatial changes in liver fibrosis.

Wanlu Liu Lab, Zhejiang University–University of Edinburgh Institute (ZJE)
Undergraduate Research Trainee, Single-Cell Transcriptomics | Jun 2025 – Mar 2026

- Assisted with single-cell RNA-seq analysis of liver samples from different disease backgrounds.
- Performed quality control, clustering, batch correction, and cell type annotation.

iZJU-China Team, International Genetically Engineered Machine (iGEM) 2025
Undergraduate Grand Prize — 2nd Runner-up (Global Top 3) | Apr 2025 – Nov 2025

- Served as a Dry Lab / Modeling Team Member. The team also received a Gold Medal, Best Infectious Diseases Project, Special Prize for Best Part Collection, and a nomination for Best Model.
- Designed a DNA origami platform for the FoCas project using cadnano, optimized and visualized structures in oxView, and carried out a stability assessment in oxDNA.
- Assisted teammates with protection assessment and transmembrane simulation, including modeling hydroxyl radical diffusion, enzyme-mediated degradation effects, and membrane-penetration behavior of the DNA origami delivery system.
- Project wiki page: Home | IZJU-CHINA - iGEM 2025

TECHNICAL SKILLS & LANGUAGES

- **Languages:** English (IELTS 7.5), Mandarin Chinese (native)
- **Programming:** Python, R, SQL, Java; basic HTML
- **Molecular & Cell Biology:** PCR, primary cell culture, AAV-mediated gene editing, immunofluorescence staining
- **Animal Experiments & Imaging:** Stereotaxic injection, oral gavage, mouse perfusion, cranial window surgery, two-photon microscopy
- **Software & Analysis Tools:** FlowJo, QuPath, Seurat, Space Ranger, cadnano, oxView, oxDNA

RELEVANT COURSEWORK

Year 1 (2024–2025)

- College Chemistry
- Advanced Mathematics
- Integrative Biomedical Sciences
- Introduction to Cellular and Molecular Biology
- Introduction to Biomedical Informatics

Year 2 (2025–2026)

- Biomedical Genetics
- Integrated Function of Body Systems
- Database and Software Technology
- Genomics and Proteomics
- Applied Data Science

ADDITIONAL INFORMATION

- **Music:** Play the pipa (traditional Chinese instrument) and guitar.
- **Interests:** Musical theatre and reading.